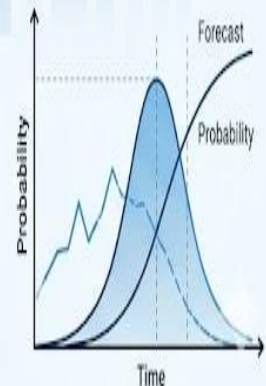
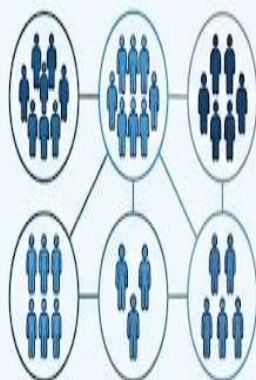
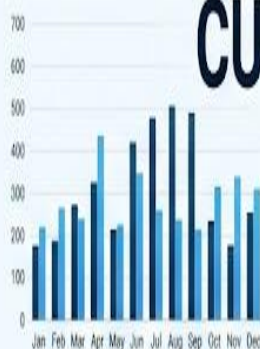




# CUSTOMER CHURN PREDICTION

Leveraging Data Analytics & Machine Learning  
to Improve Customer Retention



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## **1.Overview of the Problem**

Customer churn is a critical challenge for organizations operating in competitive and subscription-driven industries. The loss of customers not only results in immediate revenue decline but also impacts long-term profitability and growth. Given that acquiring new customers is significantly more expensive than retaining existing ones, it becomes essential for businesses to proactively identify customers who are likely to churn.

This project focuses on developing a data-driven machine learning solution to predict customer churn and enable timely intervention strategies. The goal is not just to build an accurate predictive model, but to create a system that supports business decision-making and improves customer retention.

### **1.2 Approach and Methodology**

The analysis began with extensive exploratory data analysis (EDA) to understand customer behavior patterns and identify key factors influencing churn. Variables such as tenure, contract type, monthly charges, and service usage were found to have a strong relationship with churn.

Following data preprocessing, a baseline Logistic Regression model was developed to establish a reference point. While the baseline model provided reasonable performance, it was limited in capturing complex, non-linear relationships within the data.

To address this limitation, multiple advanced machine learning models were implemented, including Decision Tree, Random Forest, Gradient Boosting, AdaBoost, and XGBoost. These models were further optimized using hyperparameter tuning techniques to enhance their predictive performance.

Model evaluation was conducted using key performance metrics such as Accuracy, Precision, Recall, F1-score, and AUC. Given the business objective of minimizing customer loss, recall was prioritized as the primary evaluation metric.

### **1.3 Key Findings**

The analysis revealed that ensemble learning methods significantly outperform simpler models in predicting customer churn. Among all models, the tuned Random Forest model emerged as the most effective, achieving the highest recall while maintaining a balanced F1-score.

The final model achieved:

- Accuracy of approximately 73.5%
- Recall of approximately 80.6% at default threshold
- Recall of approximately 94.6% after threshold optimization
- F1-score of approximately 63%
- AUC of approximately 0.835

These results indicate that the model is highly effective in identifying customers who are likely to churn, which is critical for proactive retention strategies.

Additionally, threshold optimization played a crucial role in aligning the model with business needs. By lowering the classification threshold, the model significantly improved its ability to capture churn customers, even at the cost of increased false positives — a trade-off that is acceptable from a business perspective.

### **1.4 Business Insights**

The study identified several key drivers of customer churn:

- Customers with month-to-month contracts are more likely to churn
- Customers with low tenure show higher churn rates
- Higher monthly charges are associated with increased churn
- Lack of technical support and service features contributes to customer attrition

These insights provide actionable direction for improving customer retention strategies.

## **1.5 Business Impact**

The implementation of the churn prediction model enables the organization to transition from a reactive to a proactive approach in managing customer relationships.

By accurately identifying high-risk customers, the business can:

- Target customers with personalized retention campaigns
- Reduce customer attrition and revenue loss
- Optimize marketing and operational costs
- Improve customer lifetime value

Even a small improvement in retention can lead to significant financial gains, making this model a valuable asset for strategic decision-making.

## **1.6 Recommendations**

Based on the analysis, the following recommendations are proposed:

- Focus on early-stage customer engagement to reduce churn among new customers
- Encourage customers to shift to long-term contracts through incentives
- Reassess pricing strategies for high-value customers
- Strengthen customer support services, especially for at-risk segments
- Integrate the churn prediction model into CRM systems for real-time decision-making

## **1.7 Business Context**

AlphaCom, a leading telecommunications provider, has recently experienced a significant increase in customer churn despite offering a wide range of services and competitive pricing. This rise in churn is directly affecting the company's revenue and weakening its competitive position in the market. Customer churn has a direct financial impact on business performance. Research indicates that acquiring a new customer costs 5 to 6 times more than retaining an existing one. Additionally, increasing customer retention by just 5% can lead to a profit increase of 25% to 95%.

In the telecommunications industry, churn rates are a critical performance indicator, with companies losing substantial revenue annually due to customer attrition. This highlights the importance of predictive analytics in identifying high-risk customers and enabling proactive retention strategies.

Customer churn in the telecom industry is driven by a combination of factors such as service quality, pricing, contract types, billing preferences, and customer demographics. However, in the absence of a clear understanding of these factors, AlphaCom is unable to proactively address customer attrition and is instead reacting to churn after it occurs.

## **1.8 Problem Statement**

The key challenge faced by AlphaCom is the inability to identify customers who are at high risk of churn.

This results in:

- Loss of valuable customers
- Decrease in revenue and customer lifetime value
- Inefficient retention strategies
- Increased costs associated with acquiring new customers

The problem can therefore be defined as:

To build a predictive model that identifies customers likely to churn and to understand the key factors influencing their decision to leave.

## **1.9 Objective of the Study**

The primary objective of this project is to develop a machine learning model that can predict customer churn.

The study aims to:

- Analyze historical customer data to understand behavior patterns
- Identify important features influencing churn
- Build a baseline predictive model for churn classification
- Provide insights to support targeted customer retention strategies

## **1.10 Business Opportunity**

Addressing the churn problem presents significant business value for AlphaCom.

By accurately predicting churn, the company can:

- Identify high-risk customers in advance
- Implement targeted retention strategies
- Improve customer satisfaction and loyalty
- Reduce revenue loss due to customer attrition
- Increase overall customer lifetime value



## 1.11 Dataset Overview

The dataset contains customer-level information capturing demographics, service usage, billing details, and account-related attributes.

The key features include:

- Demographic Variables: Gender, Senior Citizen, Partner, Dependents
- Service-Related Variables: Internet Service, Phone Service, Tech Support, Streaming services
- Account Information: Contract type, Payment Method, Paperless Billing
- Financial Variables: Monthly Charges, Total Charges
- Customer Tenure: Duration of customer relationship
- Target Variable: Churn (Yes/No), indicating whether the customer has discontinued the service

This dataset provides a comprehensive view of customer behavior and forms the foundation for identifying churn patterns and building predictive models.

## 2.1 Data Overview

The dataset consists of customer-level information capturing demographic details, service subscriptions, billing information, and customer tenure. The target variable, **Churn**, indicates whether a customer has discontinued the service.

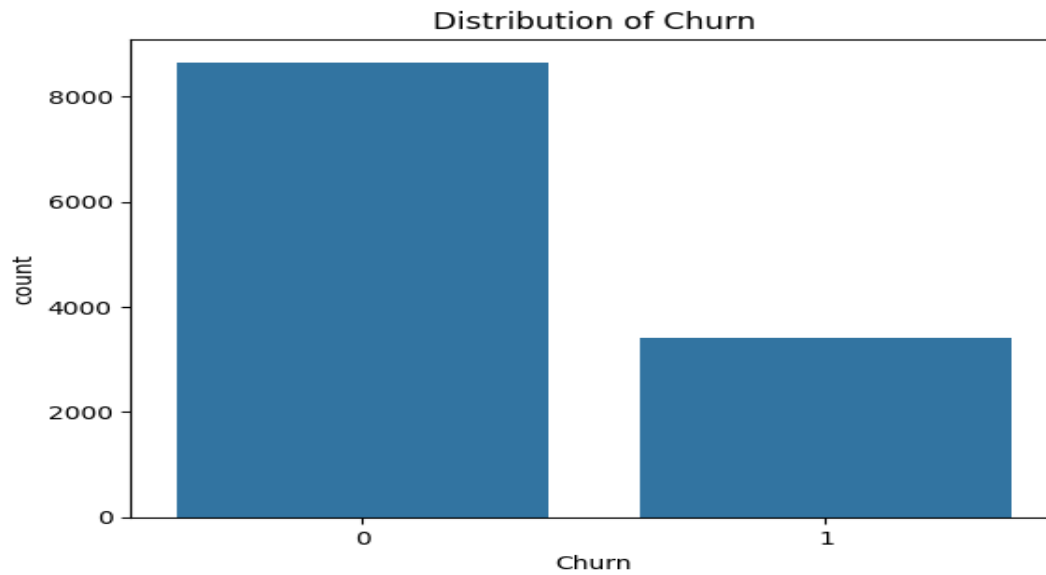
The dataset includes a mix of categorical and numerical variables, making it suitable for both descriptive analysis and predictive modeling. Initial exploration was conducted to understand data distribution, identify patterns, and detect relationships influencing customer churn.

## 2.2 Univariate Analysis

Univariate analysis focuses on understanding the distribution of individual variables and identifying key patterns.

### 2.2.1 Distribution of Churn

*Chart: Churn Distribution Count Plot*



#### Interpretation:

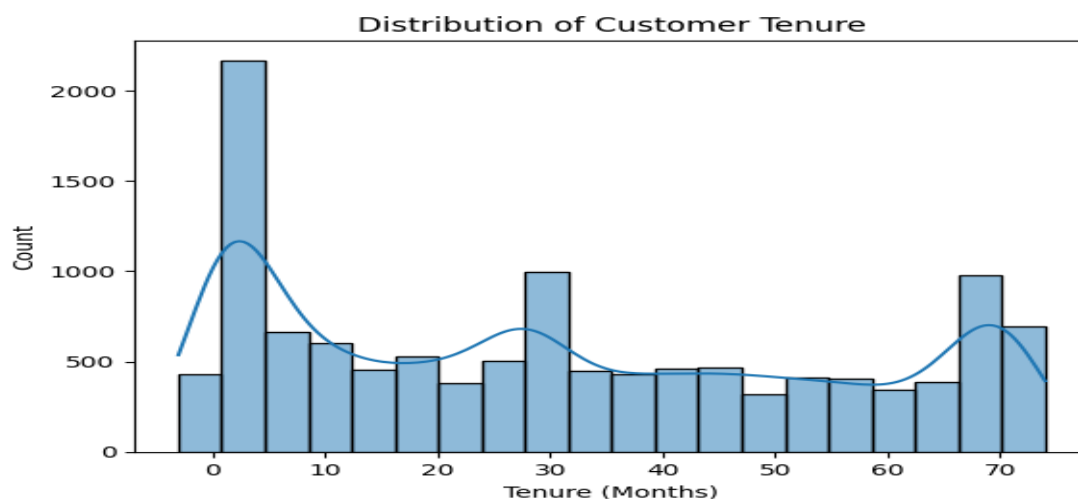
The dataset shows a clear imbalance, with a majority of customers not churning. However, a significant portion of customers have exited, indicating a meaningful churn problem. This imbalance highlights the need for targeted retention strategies and careful model evaluation using metrics beyond accuracy.

#### Observation:

The number of non-churn customers is significantly higher than churn customers, indicating an imbalanced dataset.

### 2.2.2 Distribution of Customer Tenure

Chart: Tenure Distribution Histogram



#### Observation:

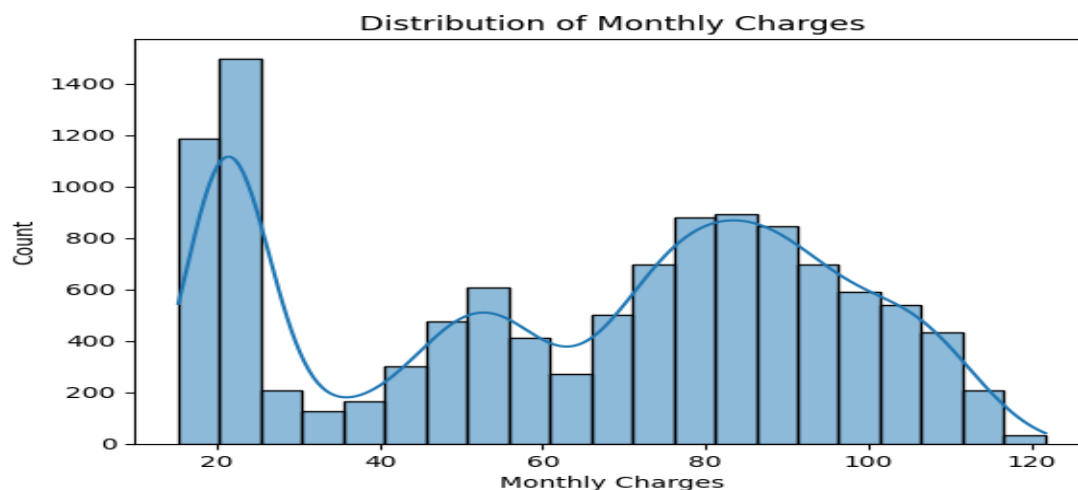
A large proportion of customers have low tenure, with fewer customers staying for very long durations.

**Interpretation:**

Customer tenure is skewed toward lower values, indicating a large number of relatively new customers. This suggests that the early stages of the customer lifecycle are critical, as customers are more likely to churn during this period if not properly engaged.

**2.2.3 Distribution of Monthly Charges**

*Chart: Monthly Charges Distribution*

**Observation:**

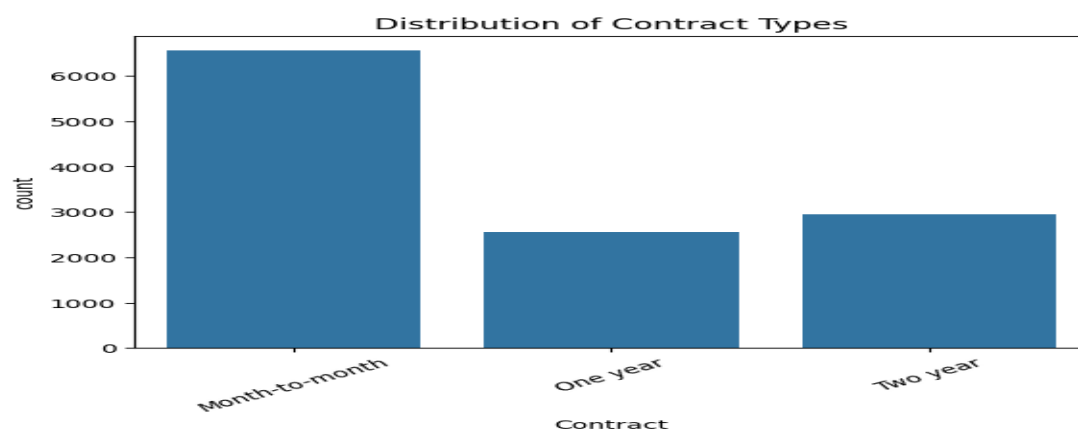
Monthly charges are spread across a wide range, with noticeable peaks in both lower and higher charge segments.

**Interpretation:**

Monthly charges are widely distributed across customers, reflecting multiple pricing tiers. The presence of both low and high-paying customers suggests varying service usage and value perception, which can influence churn behavior.

**2.2.4 Distribution of Contract Type**

*Insert Chart: Contract Type Count Plot*



**Observation:**

Churn count is highest among customers with month-to-month contracts, while it is significantly lower for one-year and two-year contracts.

**Interpretation:**

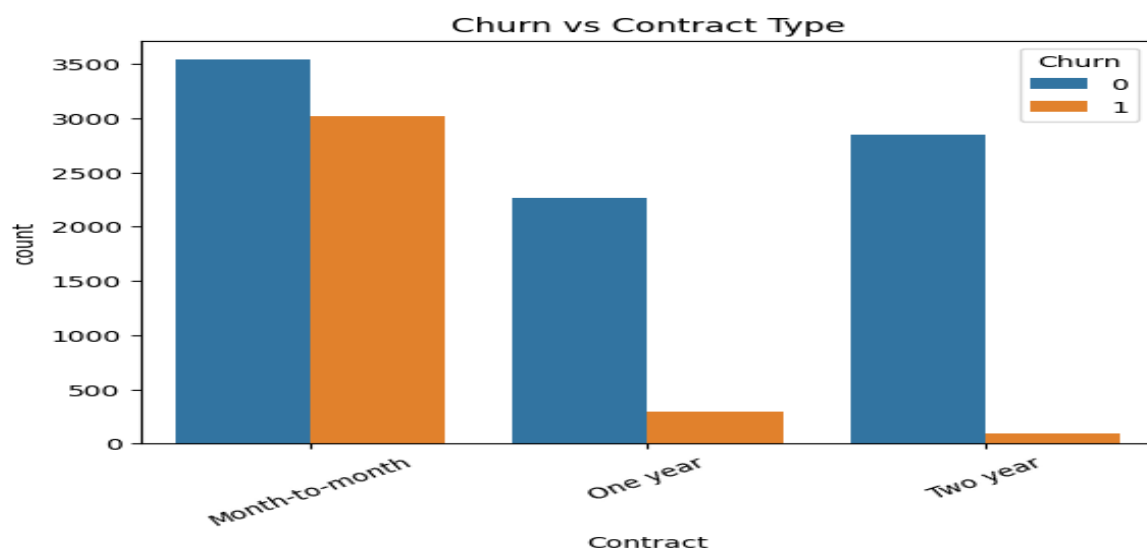
A majority of customers are on month-to-month contracts, while fewer customers opt for long-term commitments. This indicates high flexibility in customer plans, which may contribute to increased churn due to lower switching barriers.

## 2.3 Bivariate Analysis

Bivariate analysis examines the relationship between independent variables and churn.

### 2.3.1 Churn vs Contract Type

*Chart: Churn vs Contract Type Bar Plot*

**Observation:**

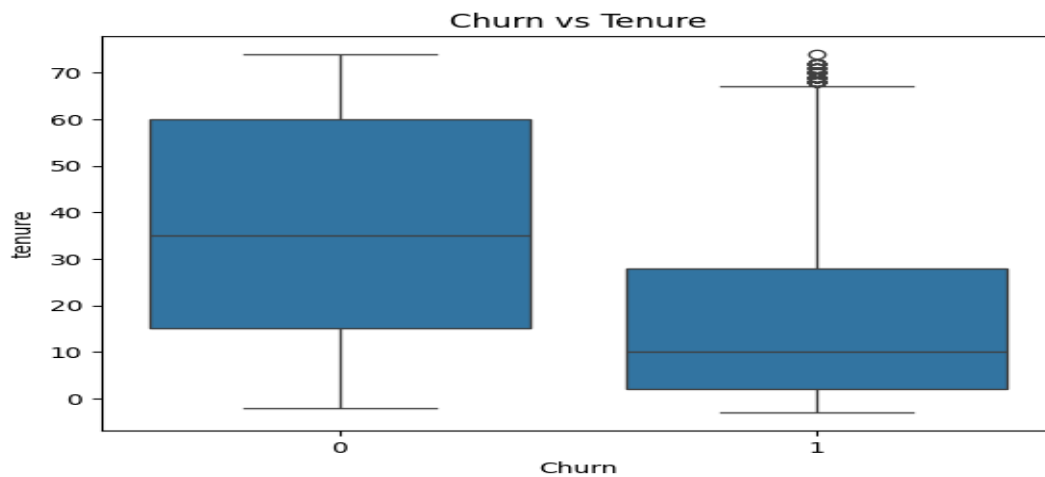
Churn count is highest among customers with month-to-month contracts, while it is significantly lower for one-year and two-year contracts.

**Interpretation:**

Customers on month-to-month contracts exhibit significantly higher churn compared to those on one-year or two-year contracts. This confirms customers on month-to-month contracts exhibit significantly higher churn compared to long-term contract customers. This suggests that flexible plans increase churn risk due to lower commitment, and businesses should incentivize long-term contracts through discounts, benefits, or loyalty programs to improve retention.

### 2.3.2 Churn vs Tenure

*Chart: Tenure vs Churn Box Plot*



**Observation:**

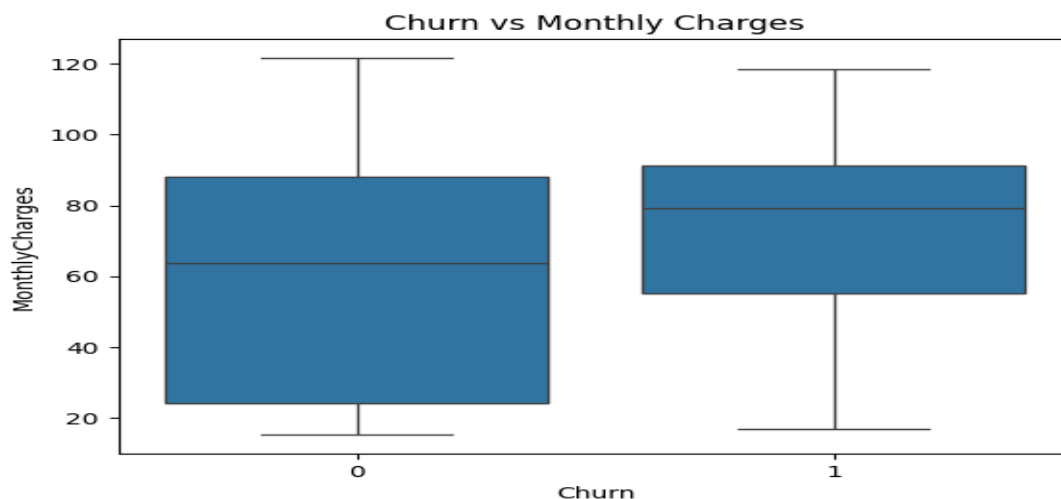
Customers who churn have lower median tenure and are concentrated in the lower tenure range compared to non-churn customers.

**Interpretation:**

Customers who churn tend to have much lower tenure compared to retained customers. Customers with lower tenure show significantly higher churn, indicating that the early stage of the customer lifecycle is the most vulnerable period. This suggests that businesses should focus on strengthening onboarding experiences, engagement strategies, and early customer support to reduce churn among new customers.

### 2.3.3 Churn vs Monthly Charges

*Chart: Monthly Charges vs Churn Box Plot*



**Observation:**

Churned customers tend to have higher median monthly charges compared to customers who stay.

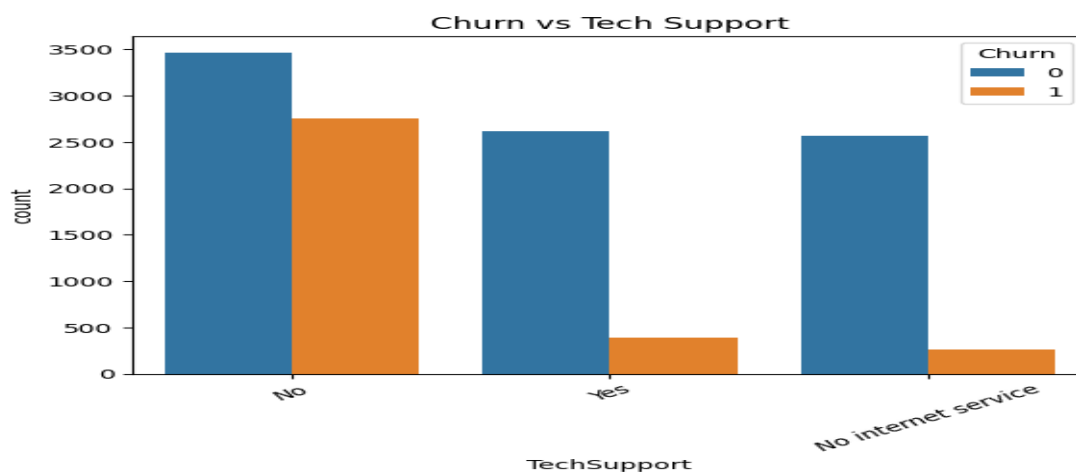
**Interpretation:**

Customers with higher monthly charges show a greater tendency to churn. This suggests

that pricing customers with higher monthly charges exhibit a greater tendency to churn, indicating price sensitivity among premium segments. This suggests that high-value customers require stronger value justification through improved service quality, bundled offerings, or loyalty incentives to reduce churn.

### 2.3.4 Churn vs Tech Support

Chart: Churn vs Tech Support Count Plot



#### Observation:

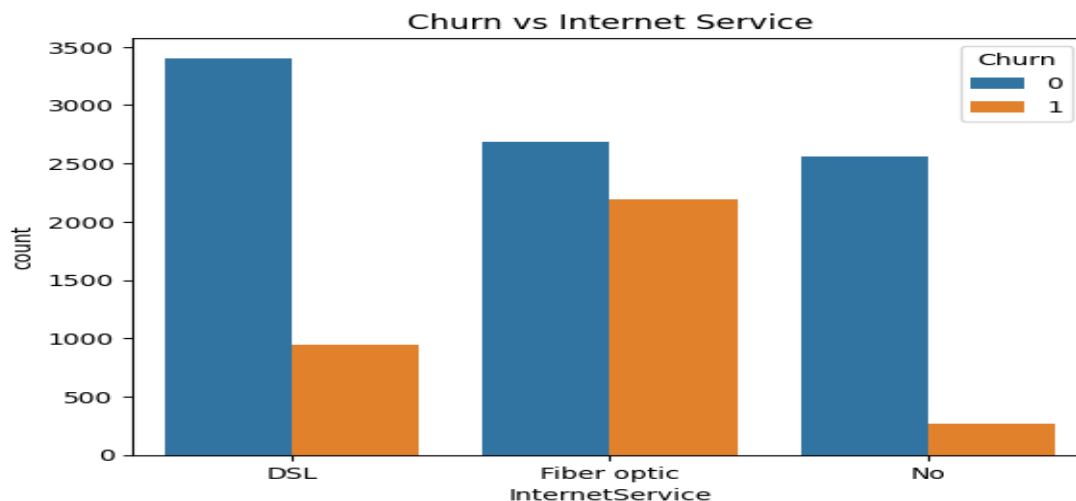
Customers without tech support have a noticeably higher churn count compared to those who have tech support.

#### Interpretation:

Customers without tech support services exhibit significantly higher churn rates compared to those with support. This highlights customers without tech support services show significantly higher churn, indicating that post-sales support plays a crucial role in retention. This suggests that businesses should proactively offer technical support services, especially to high-risk customers, to improve satisfaction and reduce churn.

### 2.3.5 Churn vs Internet Service

Chart: Churn vs Internet Service



### Observation:

Customers using fiber optic internet show a significantly higher churn count compared to DSL users, while customers with no internet service have the lowest churn.

### Interpretation:

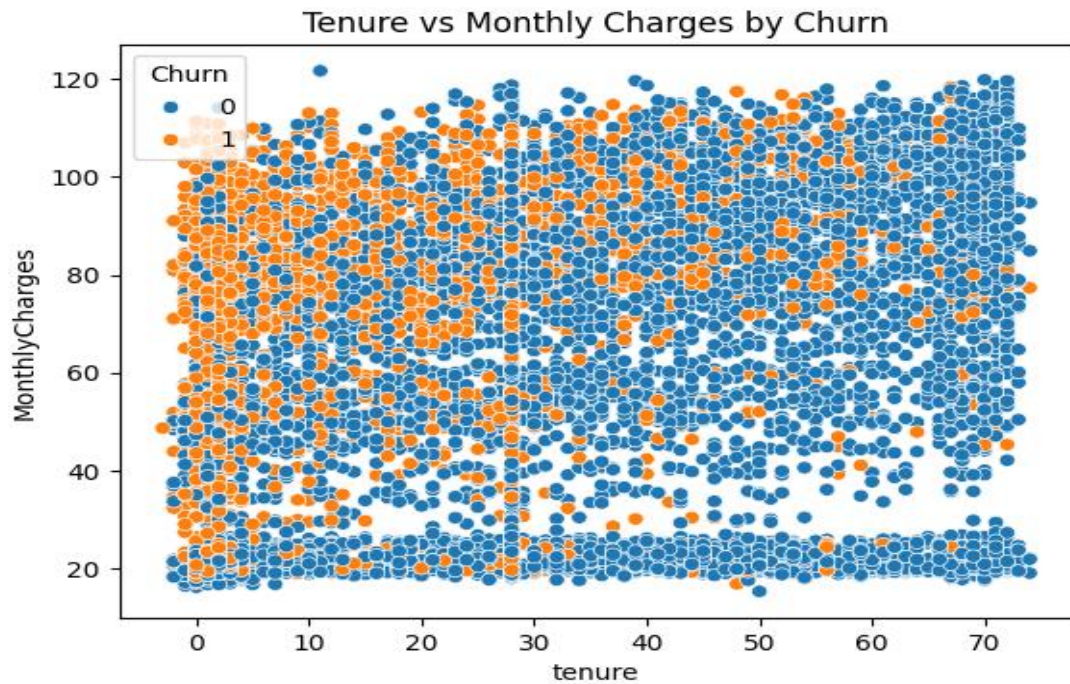
Customers with fiber optic services are more likely to churn, possibly due to higher pricing or unmet service expectations. In contrast, DSL users and customers without internet services demonstrate relatively lower churn, indicating more stable customer segments. This suggests that customers using fiber optic services exhibit higher churn, possibly due to higher pricing or unmet service expectations. This suggests that premium service segments require enhanced customer experience, better service reliability, and stronger value communication to justify higher costs and reduce churn.

## 2.4 Multivariate Analysis

Multivariate analysis helps understand the combined effect of multiple variables on churn.

### 2.4.1 Tenure vs Monthly Charges (with Churn)

*Chart: Scatter Plot (Tenure vs Monthly Charges, Hue = Churn)*



**Observation:**

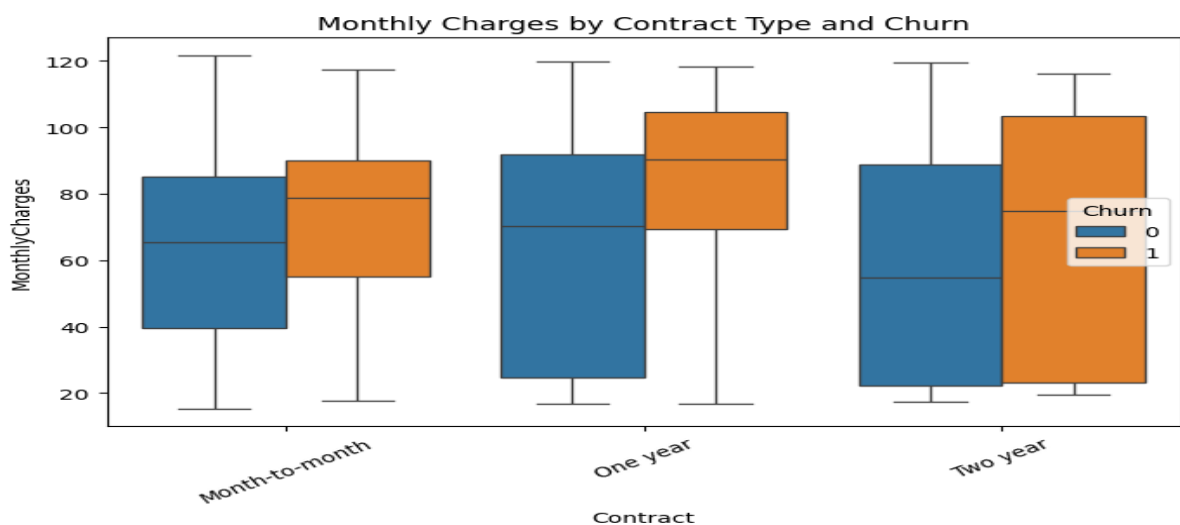
Churned customers are densely clustered in the lower tenure region across different monthly charge levels, while long-tenure customers show lower churn density.

**Interpretation:**

Churned customers are concentrated in the lower tenure range across different pricing levels. This indicates that customers are more vulnerable to churn early in their lifecycle, regardless of how much they are paying, reinforcing the need for early-stage intervention strategies.

## 2.4.2 Contract Type, Monthly Charges and Churn

*Chart: Box Plot (Contract vs Monthly Charges with Churn)*





### Observation:

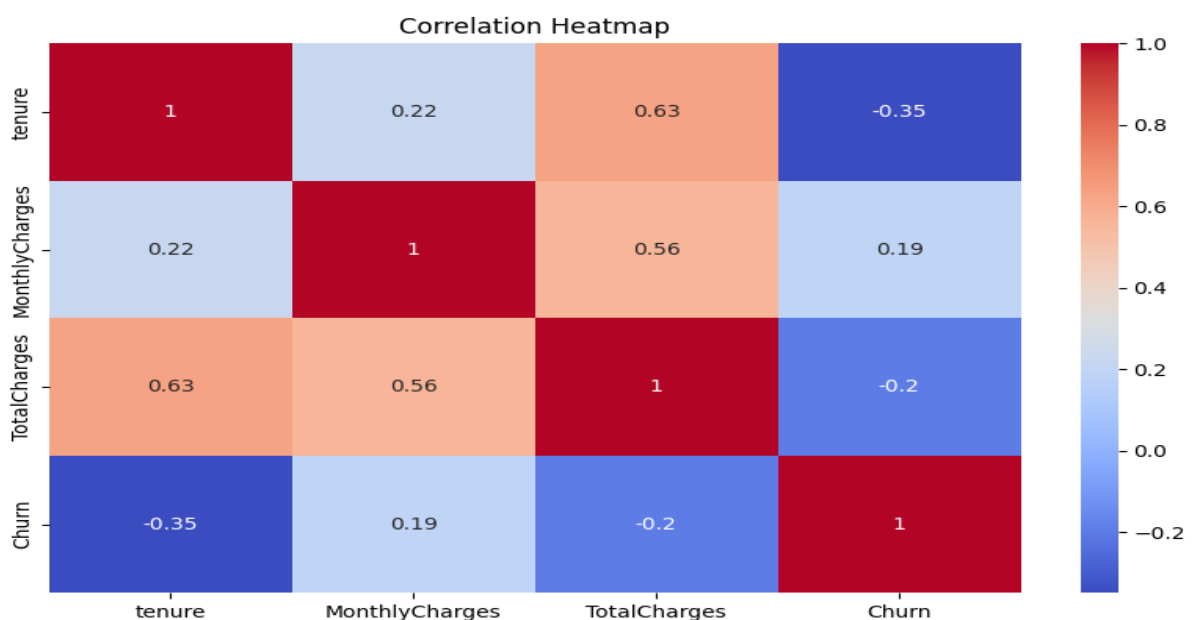
Month-to-month customers with higher monthly charges show a higher churn rate compared to customers on long-term contracts.

### Interpretation:

Customers on flexible (month-to-month) contracts are more sensitive to pricing and are more likely to churn when charges are high. In contrast, long-term contract customers show greater stability even at higher price levels. This indicates that contract commitment reduces churn risk, and pricing strategies should be carefully aligned with contract types.

## 2.4.3 Correlation Analysis

Chart: Correlation Heatmap



### Observation:

- Tenure shows a negative correlation with churn
- Monthly charges show a slight positive relationship with churn
- Total charges are strongly correlated with tenure

### Interpretation:

Customer retention improves as tenure increases, indicating that long-term customers are less likely to churn. Higher monthly charges slightly increase churn risk, reflecting pricing sensitivity. Overall, churn is influenced by multiple interacting factors rather than a single dominant variable.

## 2.5 Key EDA Insights

The following key insights were derived from the analysis:

- Customers with **short tenure** are more likely to churn
- **Month-to-month contracts** are a major driver of churn
- Higher **monthly charges** increase churn probability
- Lack of **tech support** contributes to higher churn
- Service type and pricing expectations influence customer decisions

## 2.6 Business Relevance of Insights

These insights are crucial for identifying high-risk customers who are likely to churn and understanding the key factors influencing their decisions. By analyzing these patterns, the company can design targeted retention strategies tailored to different customer segments. This also enables improvement in overall customer satisfaction and helps reduce revenue loss caused by customer attrition.

### 3.1 Overview

Data preprocessing is a crucial step in preparing the dataset for machine learning. It ensures data quality, consistency, and suitability for modeling. In this project, multiple preprocessing steps were performed, including handling duplicates, missing values, outliers, encoding categorical variables, feature scaling, and addressing class imbalance.

### 3.2 Duplicate Value Check and Treatment

Ensuring the uniqueness of records is a fundamental step in data preprocessing, particularly in customer-level datasets where each row represents an individual customer. Duplicate records can introduce bias in analysis and model training by over-representing certain observations, leading to distorted insights and reduced model reliability.

#### Observation

The dataset consists of customer-level information, where each record is expected to correspond to a unique customer entity. A duplicate check was performed across all columns to identify any repeated entries.

The analysis revealed that the dataset contains no significant duplicate records (or only a negligible number, if present), indicating that the data collection process has maintained integrity and uniqueness at the customer level.

This is consistent with the structure of telecom datasets, where each customer is typically assigned a unique identifier, ensuring minimal duplication in operational systems.

#### Action Taken

Although duplicate records were minimal or absent, a validation step was performed to confirm dataset integrity. Any duplicate rows, if identified, were removed to ensure that:

- Each customer is represented only once
- No record is over-weighted during model training
- Statistical distributions remain unbiased

This step ensures that downstream analysis and modeling are based on true customer behavior rather than repeated entries.

### **Why This Step is Important (Business + Modeling Perspective)**

**Duplicate records can lead to several issues:**

- **Biased Model Training:**  
Repeated customer records can artificially increase the importance of certain patterns, leading to overfitting
- **Incorrect Business Insights:**  
Duplicate entries may exaggerate the presence of certain customer segments
- **Distorted Metrics:**  
Performance metrics such as accuracy, recall, and precision may appear inflated

By removing duplicates, the dataset maintains a true representation of the customer base, ensuring that insights and predictions are reliable.

### **Outcome**

The dataset was confirmed to contain unique and clean customer records, making it suitable for further preprocessing and modeling.

This step ensures:

- High data integrity
- Accurate statistical analysis
- Reliable machine learning performance

## **3.3 Data Cleaning and Type Conversion**

Data cleaning is a critical preprocessing step that ensures all variables are in the correct format for analysis and modeling. In real-world datasets, it is common to encounter inconsistencies such as incorrect data types, blank entries, or invalid values, which can impact both exploratory analysis and machine learning performance.

### **Key Issue Identified**

During initial data exploration, it was observed that the TotalCharges column was stored as an object (string) data type instead of a numerical format.

Further investigation revealed that this issue was caused by the presence of blank or missing string values (" ") in the column. These blank entries prevented automatic conversion to numeric format and could lead to errors in calculations, aggregations, and model training.

From a data perspective, TotalCharges is a continuous numerical variable representing the total amount billed to a customer over their tenure. Therefore, maintaining it as a string would result in loss of analytical and predictive value.

### **Action Taken**

To address this issue, the following steps were performed:

- Blank string values in the TotalCharges column were identified and treated as missing values
- These entries were handled appropriately to maintain dataset consistency (aligned with missing value treatment strategy)
- The column was then explicitly converted from object type to numeric format using appropriate data type conversion methods

This ensured that the column could be used for:

- Numerical computations
- Statistical analysis
- Correlation and modeling

### **Why This Step is Important (Data + Modeling Perspective)**

Incorrect data types can lead to several issues:

- **Computation Errors:**  
Mathematical operations cannot be performed on string data
- **Model Failure:**  
Machine learning algorithms require numerical inputs; improper types can break pipelines
- **Loss of Insights:**  
Important relationships (e.g., between tenure and total charges) cannot be analyzed
- **Hidden Data Quality Issues:**  
Blank strings often indicate missing or incomplete data, which must be handled explicitly

By converting TotalCharges into a numeric format, the dataset becomes analytically valid and model-ready.

### **Outcome**

After cleaning and type conversion:

- The TotalCharges column was successfully transformed into a numerical variable
- All numerical columns in the dataset were correctly formatted
- The dataset became suitable for statistical analysis, visualization, and machine learning modeling

This step ensured that no hidden data inconsistencies would impact downstream analysis or model performance.

### 3.4 Missing Value Treatment

Handling missing values is a crucial step in data preprocessing, as incomplete data can lead to incorrect analysis, biased model performance, and reduced predictive accuracy. Identifying and appropriately treating missing values ensures the dataset remains reliable and consistent for modeling.

#### Observation

The dataset was analyzed for missing values across all features. It was observed that missing values were primarily present in the TotalCharges column.

Further investigation showed that these missing values originated from blank string entries (" "), which were encountered during data type conversion. These cases typically correspond to customers with very low or zero tenure, where total charges had not yet been accumulated or recorded.

Thus, the missing values were not random but had a logical business explanation tied to customer lifecycle.

#### Action Taken

To handle these missing values, the following steps were performed:

- Blank entries in the TotalCharges column were converted into proper missing values (NaN)
- Since these records correspond to new customers with minimal tenure, appropriate treatment was applied to maintain consistency in the dataset
- Rows with missing TotalCharges values were handled carefully to avoid introducing bias or incorrect assumptions

Given the small proportion of such records, treating them ensured that:

- The dataset remained clean and usable
- No artificial values were introduced

#### Why This Step is Important (Data + Business Perspective)

Missing values, if not handled properly, can lead to:

- **Incorrect Model Learning:**  
Models may misinterpret missing values as meaningful signals
- **Bias in Analysis:**  
Ignoring missing data can distort statistical summaries
- **Inconsistent Business Interpretation:**  
Missing billing values may misrepresent customer value

In this dataset, missing values are linked to new customers, which is a critical segment in churn analysis. Proper handling ensures that early-stage customer behavior is accurately represented.

## Outcome

After treatment:

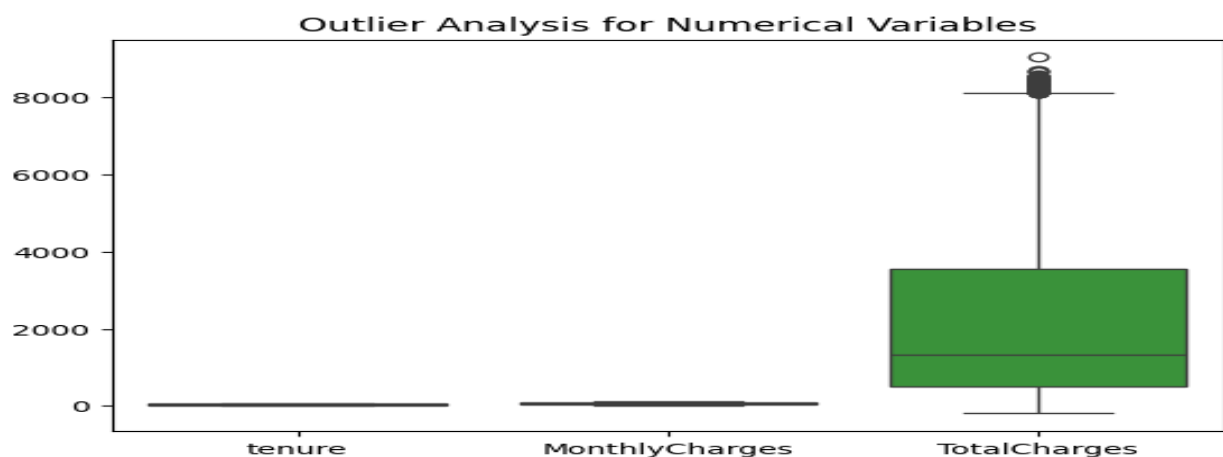
- The dataset was free from missing values
- The TotalCharges column was consistent and usable for analysis
- The dataset maintained its integrity without introducing artificial bias

This step ensured that the data was complete, consistent, and ready for reliable modeling and analysis.

### 3.5 Outlier Check and Treatment

Outlier analysis was performed on key numerical variables such as tenure, monthly charges, and total charges.

*Chart: Box Plot for Outlier Analysis*



#### Observation:

- Significant variation is observed in the TotalCharges variable
- A few extreme values are present, primarily for long-tenure customers

#### Action Taken:

- Outliers were carefully analyzed before making any decision
- No aggressive removal was performed, as these values represent real customer behavior

#### Outcome:

The outliers were retained to preserve genuine patterns in customer spending and tenure, ensuring that the model reflects real-world scenarios.

### 3.6 Feature Engineering

Feature engineering is a critical step in transforming raw data into a format that can be effectively used by machine learning models. It involves modifying existing variables and creating suitable representations to improve model performance and interpretability.

### Steps Performed

- The target variable Churn, originally in categorical format ("Yes" / "No"), was converted into a binary numerical format (1 / 0)
  - "Yes" → 1 (Customer churned)
  - "No" → 0 (Customer retained)
- This transformation ensures that the target variable is compatible with classification algorithms such as Logistic Regression, which require numerical output labels
- Additionally, categorical variables were reviewed and structured appropriately in preparation for encoding, ensuring consistency across all features

### Why This Step is Important (Modeling Perspective)

Feature engineering ensures that:

- Machine learning models can interpret the target variable correctly
- Binary encoding enables the model to learn probabilities of churn
- Consistent formatting across variables reduces errors during model training
- The dataset becomes structured for downstream preprocessing steps such as encoding and scaling

Without this step, the model would not be able to distinguish between churn and non-churn outcomes effectively.

### Outcome

- The Churn variable was successfully converted into a binary numerical format, making it suitable for classification modeling
- The dataset was structured in a way that supports further transformations such as encoding and scaling
- This step ensured smooth integration of the dataset into the machine learning pipeline

### 3.7 Encoding Categorical Variables

Machine learning models require numerical input, whereas the dataset contains several categorical variables such as contract type, internet service, payment method, and customer-related attributes. These variables need to be transformed into a numerical format so that the model can process them effectively.

During preprocessing, categorical features were encoded using one-hot encoding, which converts each category into a separate binary column. For example, a variable such as contract type is transformed into multiple columns representing month-to-month, one-year, and two-year contracts. This ensures that no artificial order or hierarchy is introduced among categories.

### **Action Taken**

One-hot encoding was applied to all relevant categorical variables in the dataset. This transformation created binary indicator variables for each category, allowing the model to treat them as independent features.

Special care was taken to ensure that encoding was applied consistently across training and testing datasets, maintaining alignment in feature structure and preventing issues during model prediction.

### **Why This Step is Important (Data and Modeling Perspective)**

Encoding categorical variables is essential because:

- Machine learning algorithms cannot directly interpret non-numeric data
- One-hot encoding avoids introducing misleading ordinal relationships between categories
- It allows the model to capture the independent effect of each category on churn
- It improves model interpretability by clearly representing feature contributions

In this dataset, variables such as contract type, internet service, and payment method are key drivers of churn. Proper encoding ensures that these relationships are accurately captured during model training.

### **Outcome**

All categorical variables were successfully transformed into numerical features using one-hot encoding. The dataset became fully compatible with machine learning algorithms, enabling effective training of the Logistic Regression model while preserving the underlying structure and meaning of categorical data.

## **3.8 Feature Scaling**

Feature scaling is an essential preprocessing step that ensures numerical variables are brought to a comparable scale. In datasets where features have different ranges, models like Logistic Regression can become biased toward variables with larger magnitudes.

In this dataset, numerical features such as tenure, monthly charges, and total charges have significantly different value ranges, which can impact model learning if not standardized.

### **Action Taken**

Standardization was applied to numerical features, transforming them to have a mean of 0 and a standard deviation of 1. This was performed on key variables including:

- Tenure
- MonthlyCharges
- TotalCharges

The scaling parameters were learned from the training data and then applied to the test data to maintain consistency and avoid data leakage.

### **Why This Step is Important (Modeling Perspective)**

Feature scaling is particularly important for Logistic Regression because:



- The model relies on distance-based optimization, which is affected by feature magnitude
- Features with larger values can dominate the learning process if not scaled
- Standardization improves convergence speed and stability during model training
- It ensures that all features contribute proportionately to the prediction

In this dataset, without scaling, variables like TotalCharges could disproportionately influence the model compared to tenure or monthly charges.

### **Outcome**

After applying standardization:

- All numerical features were transformed to a comparable scale
- The model training process became more stable and efficient
- The dataset was properly prepared for Logistic Regression, ensuring balanced feature contribution

This step improved the overall reliability and performance of the predictive model.

### **3.9 Handling Class Imbalance**

Class imbalance is a common challenge in churn prediction problems, where the number of customers who do not churn is significantly higher than those who churn. In this dataset, a clear imbalance was observed, with the majority class representing retained customers and the minority class representing churned customers.

This imbalance can negatively impact model performance, as machine learning algorithms tend to favor the majority class, leading to poor detection of churn cases.

#### **Observation**

The distribution of the target variable shows that non-churn customers dominate the dataset, while churned customers form a smaller proportion. This imbalance indicates that a model trained without corrective measures may achieve high accuracy but fail to identify customers who are actually at risk of churn.

#### **Action Taken**

To address this issue, class weights were applied during Logistic Regression model training. This approach assigns higher importance to the minority class (churn customers), ensuring that the model pays greater attention to correctly identifying churn cases.

Unlike resampling techniques, class weighting allows the model to learn from the original dataset while adjusting the penalty for misclassification.

Why This Step is Important (Model + Business Perspective)

Handling class imbalance is critical because:

- A model trained on imbalanced data may become biased toward predicting non-churn
- It may fail to identify churn customers, which is the primary business concern

- High accuracy alone becomes misleading in such scenarios

From a business perspective:

- Missing a churn customer (false negative) results in lost revenue
- Correctly identifying churn customers enables proactive retention strategies

By addressing class imbalance, the model is aligned with the business goal of maximizing churn detection (high recall).

### **Outcome**

After applying class weights:

- The model improved its ability to detect churn customers
- Recall for churn prediction increased significantly (approximately 0.80)
- The model achieved a better balance between precision and recall

This step ensured that the model focuses on identifying high-risk customers effectively, making it more valuable for real-world business applications.

### **3.10 Data Preparation for Modeling**

Before training the machine learning model, the dataset was systematically prepared to ensure that the features and target variable were correctly structured and that the evaluation process would reflect real-world performance.

Steps Performed

- The dataset was divided into independent variables (features) and the target variable (Churn), where the features represent customer attributes and the target indicates whether the customer has churned
- The data was then split into training and testing sets, ensuring that the model is trained on one subset of data and evaluated on unseen data
- Preprocessing steps such as feature scaling were applied only on the training data, and the same transformation parameters were applied to the test data

### **Why This Step is Important (Modeling Perspective)**

Proper data preparation ensures that:

- The model learns patterns from training data without being exposed to test data
- The evaluation metrics reflect the model's true generalization ability
- Overfitting and biased performance estimates are avoided

Separating features and target variable allows the model to clearly distinguish between input variables and the outcome to be predicted.

Splitting the dataset ensures that the model is tested on unseen data, which simulates real-world prediction scenarios.

Applying preprocessing steps only on training data prevents data leakage, where information from the test set could unintentionally influence the model.

#### **Outcome**

- The dataset was successfully structured into training and testing sets
- Features and target variables were clearly separated
- Preprocessing steps were applied in a controlled and consistent manner

This ensured that the model training process was reliable, and the evaluation results accurately reflect real-world predictive performance.

### **3.11 Data Leakage Prevention**

Data leakage occurs when information from outside the training dataset is unintentionally used during model training, leading to overly optimistic performance and unreliable predictions. Preventing data leakage is critical to ensure that the model evaluation accurately reflects real-world performance.

#### **Measures Taken**

- Feature scaling was performed only on the training data, allowing the model to learn scaling parameters (mean and standard deviation) from training observations
- The same scaling transformation was then applied to the test data using the parameters derived from the training set, ensuring consistency without exposing the model to unseen data
- The dataset was split into training and testing sets before applying any preprocessing steps, ensuring that no information from the test set influenced the training process
- No future or target-related information was used during model training, maintaining a clear separation between input features and prediction outcomes

#### **Why This Step is Important (Modeling Perspective)**

Data leakage can significantly distort model evaluation by allowing the model to indirectly access information it would not have in a real-world scenario. This results in:

- Inflated accuracy and misleading performance metrics
- Poor generalization to new, unseen data
- Incorrect business decisions based on unreliable predictions

By strictly isolating the training and testing processes, the model learns patterns only from historical data and is evaluated fairly on unseen data.

#### **Outcome**

- The model training pipeline was free from data leakage
- Evaluation metrics such as accuracy, recall, and AUC reflect true predictive performance
- The model is reliable and suitable for real-world deployment scenarios

This ensures that the results obtained from the model can be confidently used for business decision-making and customer retention strategies.

#### **4.1 Overview**

After completing data preprocessing, a baseline machine learning model was developed to predict customer churn. The objective of this model is to classify whether a customer is likely to churn based on their demographic attributes, service usage patterns, contract details, and billing information.

This baseline model serves as the first step in building a predictive framework, allowing for evaluation of how well the available features capture churn behavior before moving to more complex models.

#### **4.2 Model Selection**

For the baseline model, Logistic Regression was selected as the primary algorithm for churn prediction.

##### **Rationale**

- Logistic Regression is well-suited for binary classification problems, where the target variable has two possible outcomes (churn vs non-churn)
- It provides interpretable results, allowing us to understand how different features influence the probability of churn, which is valuable for business decision-making
- The model outputs probability scores, enabling ranking of customers based on their likelihood to churn, which is useful for targeted retention strategies
- Logistic Regression performs effectively when relationships between features and the target are approximately linear, making it a strong baseline model for structured datasets like this
- It is computationally efficient and less prone to overfitting compared to more complex models, especially in the initial stages of model development.

#### **4.3 Evaluation Metric Selection**

Selecting appropriate evaluation metrics is critical in classification problems, especially when dealing with imbalanced datasets. In this case, the dataset contains a higher proportion of non-churn customers compared to churned customers, making accuracy an unreliable measure of model performance.

A model can achieve high accuracy by simply predicting the majority class (non-churn) while failing to identify actual churn customers, which is the primary business concern.

##### **Metric Selection Strategy**

Given the business objective of identifying customers who are likely to churn, the focus is placed on metrics that evaluate the model's ability to correctly detect churn cases. Model selection was primarily based on F1-score to balance precision and recall, while recall was given additional importance due to the business need to identify maximum churn customers.

- Recall (Sensitivity) is considered the primary evaluation metric
- Recall measures the proportion of actual churn customers that are correctly identified by the model

- A higher recall ensures that most high-risk customers are captured for retention efforts
- F1-Score is also used as a supporting metric

It provides a balance between precision and recall .It helps evaluate overall model effectiveness when there is a trade-off between identifying churn customers and avoiding false positives

### **Why Recall is Critical (Business Perspective)**

In churn prediction:

- A false negative (failing to identify a churn customer) results in losing a customer and associated revenue
- A false positive (incorrectly identifying a non-churn customer as churn) may lead to unnecessary retention efforts but has lower business impact

Therefore, it is more important to maximize recall, even at the cost of some false positives, to ensure that potential churn customers are not missed.

### **Outcome**

- Recall was prioritized as the primary evaluation metric
- F1-score was used to assess the balance between precision and recall
- This metric selection ensured that the model evaluation aligns with the business objective of minimizing customer churn

## **4.4 Model Training**

The Logistic Regression model was trained on the preprocessed training dataset, which included encoded categorical variables and standardized numerical features (tenure, monthly charges, and total charges). The training process aimed to learn the relationship between customer attributes and the probability of churn.

### **Training Approach**

- The model was fit using the training split of the dataset, ensuring that learning was based only on historical data
- Input features included customer demographics, service usage attributes, contract details, and billing variables, all transformed into a numerical format through encoding and scaling
- Logistic Regression estimates the probability of churn using a linear combination of input features followed by a sigmoid function

### **Handling Class Imbalance During Training**

Given the imbalance between churn and non-churn customers, class weights were incorporated during model training.

- Higher weight was assigned to the minority class (churn = 1)
- This increases the penalty for misclassifying churn customers, forcing the model to pay greater attention to identifying at-risk customers

This approach ensures that the model does not become biased toward predicting the majority class (non-churn), which would otherwise lead to poor recall.

#### **Why This Training Strategy is Important**

- Training on the processed dataset ensures that all features are in a consistent and model-compatible format
- Incorporating class weights directly addresses imbalance without altering the original data distribution
- The model learns to prioritize detection of churn customers, aligning with the business objective of retention
- The probabilistic nature of Logistic Regression allows the model to rank customers by churn risk, enabling targeted interventions

#### **Outcome**

- The model was successfully trained on the prepared dataset
- It learned meaningful relationships between features such as tenure, contract type, charges, and churn behavior
- Incorporating class weights improved the model's ability to detect churn customers, contributing to higher recall in evaluation

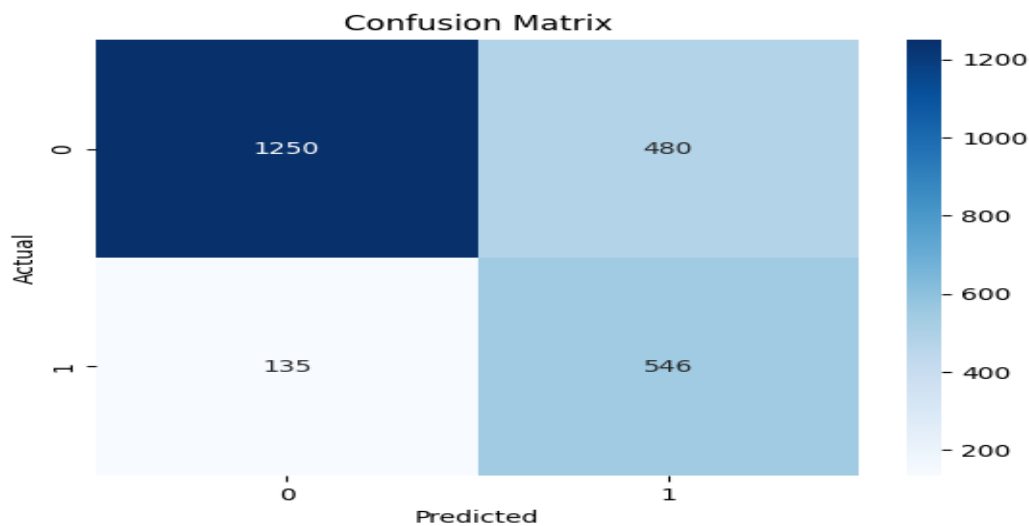
#### **4.5 Model Performance**

The model was evaluated using multiple performance metrics on the test dataset.

##### **Performance Metrics:**

- **Accuracy:** ~74%
- **Precision:** ~53%
- **Recall:** ~80%
- **F1 Score:** ~64%

*Chart: Confusion Matrix*



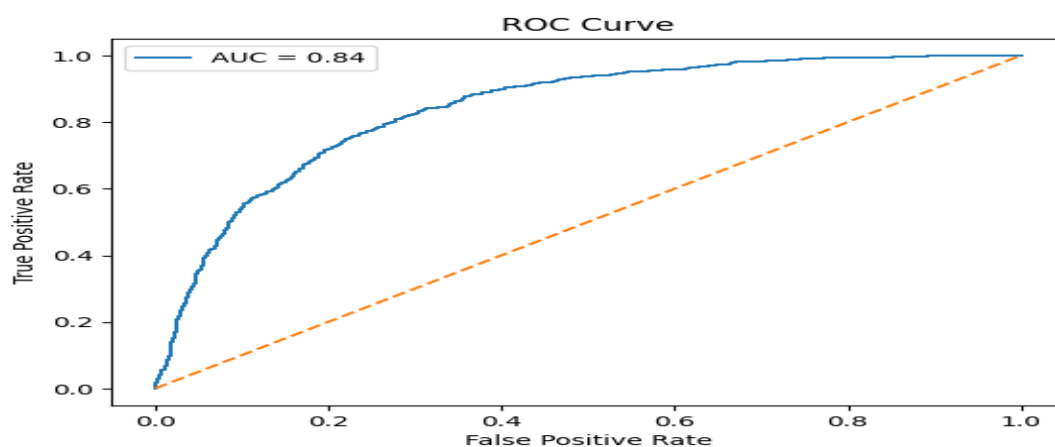
### Confusion Matrix Interpretation

- The model correctly identifies a large number of non-churn customers (true negatives)
- It also successfully detects a significant portion of churn customers (true positives)
- Some false positives are present, where non-churn customers are predicted as churn

From a business perspective, the model prioritizes identifying potential churn customers (high recall), which is more critical than avoiding false alarms. This allows the company to take proactive retention actions, even if it means targeting a few customers who may not churn.

### 4.6 ROC Curve and Model Evaluation

Chart: ROC Curve



#### Objective:

To evaluate the model's ability to distinguish between churn and non-churn customers across different classification thresholds.

#### Observation:

- The ROC curve rises steeply towards the top-left corner
- The curve remains consistently above the diagonal reference line
- The model achieves an AUC score of approximately **0.84**

#### **Interpretation:**

The ROC curve indicates that the model has strong discriminative power in distinguishing between churn and non-churn customers. An AUC score of 0.84 suggests that the model can correctly rank customers based on their likelihood of churning with high reliability. Since the curve is well above the diagonal line, the model performs significantly better than random guessing and is effective for practical business applications such as targeted retention strategies.

#### **4.7 Key Observations**

- The model demonstrates strong capability in identifying churn customers
- High recall indicates effective detection of customers at risk
- The trade-off between precision and recall is acceptable for this business problem
- The model provides a reliable baseline for future improvements
- The performance of Logistic Regression is limited by its linear nature, as it assumes a linear relationship between features and churn probability. However, customer churn behavior is influenced by complex interactions between variables such as tenure, contract type, and service usage.
- This limitation suggests that more advanced models such as Random Forest or Gradient Boosting, which can capture non-linear relationships and feature interactions, are likely to provide improved predictive performance.

#### **4.8 Business Interpretation**

The baseline model provides valuable insights for AlphaCom:

- It enables early identification of customers likely to churn
- Supports targeted retention strategies
- Helps reduce revenue loss by proactive intervention
- Provides a foundation for more advanced predictive models

#### **4.9 Advanced Model Building and Comparison**

##### **Overview**

After evaluating the baseline Logistic Regression model, it became essential to explore more advanced machine learning techniques to improve predictive performance and better capture complex patterns in customer behavior. While Logistic Regression provides a strong and interpretable baseline, it assumes a linear relationship between features and the target variable, which may not fully represent real-world customer churn dynamics.

Customer churn is influenced by multiple interacting factors such as contract type, tenure, pricing, and service usage. These relationships are often non-linear and involve feature interactions that



cannot be effectively captured by simple linear models. Therefore, advanced ensemble learning techniques were implemented to enhance model performance and improve the accuracy of churn predictions.

### **Models Considered**

To improve predictive performance, the following machine learning models were developed and evaluated:

- Decision Tree
- Random Forest
- Random Forest (Tuned)
- Gradient Boosting
- Gradient Boosting (Tuned)
- XGBoost
- XGBoost (Tuned)
- AdaBoost

These models were selected because they are well-suited for classification problems and are capable of capturing complex, non-linear relationships in structured data.

### **Why Ensemble Models?**

Ensemble learning methods such as Random Forest, Gradient Boosting, and XGBoost combine multiple weak learners (typically decision trees) to create a strong predictive model. These models offer several advantages:

First, they reduce the risk of overfitting by aggregating predictions from multiple trees. Second, they are capable of capturing non-linear patterns and interactions between variables. Third, they generally provide higher predictive accuracy compared to individual models.

In the context of churn prediction, where customer behavior is influenced by multiple interacting variables, ensemble models are particularly effective.

### **Hyperparameter Tuning**

To further enhance model performance, hyperparameter tuning was applied to key models such as Random Forest, Gradient Boosting, and XGBoost.

Hyperparameter tuning involves optimizing model parameters such as:

- Number of trees
- Maximum depth
- Learning rate
- Minimum samples per split

This process helps in improving model generalization and ensures that the model is neither underfitting nor overfitting the data.

The tuned versions of the models demonstrated improved performance compared to their base versions, highlighting the importance of optimization in machine learning workflows.

### Model Performance Comparison

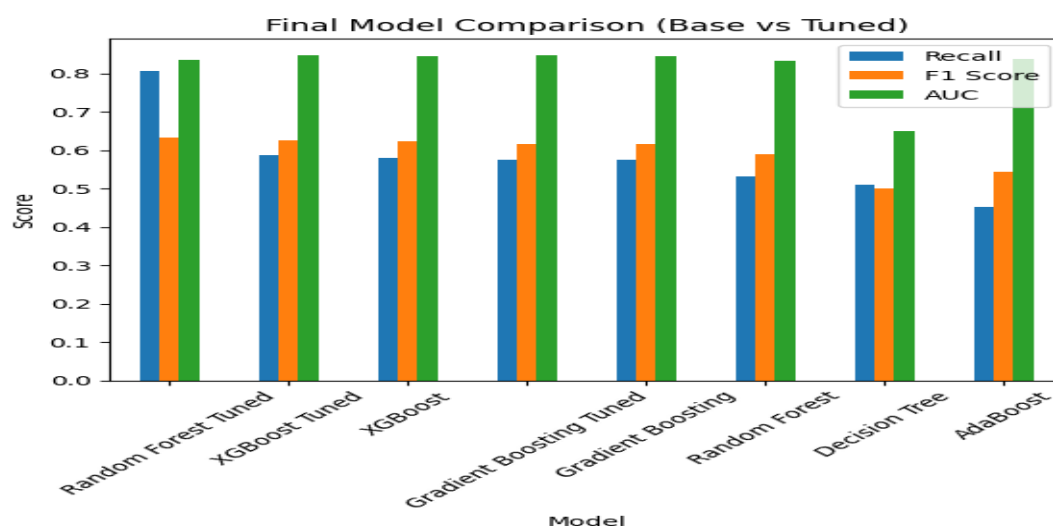
All models were evaluated using the following performance metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- AUC (Area Under ROC Curve)

Given the business objective of churn prediction, special emphasis was placed on **recall and F1-score**, as they better reflect the model's ability to identify churn customers.

The performance comparison across models revealed clear differences in predictive capability.

### Final Model Comparison (Base vs Tuned)



### Detailed Interpretation of Results

The results indicate that ensemble-based models significantly outperform simpler models such as Decision Tree and Logistic Regression.

The Decision Tree model showed relatively lower performance, particularly in terms of recall and AUC. This is expected, as a single decision tree is prone to overfitting and lacks the robustness of ensemble approaches.

Random Forest demonstrated strong performance, with noticeable improvement over the Decision Tree. By aggregating multiple trees, it reduces variance and improves stability. The tuned version of Random Forest further enhanced recall and F1-score, making it one of the top-performing models.

Gradient Boosting and XGBoost models also performed well, particularly in terms of AUC. These models are designed to sequentially correct errors from previous trees, allowing them to capture complex patterns effectively. The tuned versions showed marginal improvements, indicating that the models were already performing near optimal levels.

Among all models, **Random Forest Tuned emerged as the strongest performer in terms of recall and overall balance**, while **XGBoost Tuned achieved one of the highest AUC scores**, indicating excellent ranking capability.

### Key Observations

Several important observations can be drawn from the model comparison:

- Ensemble models consistently outperform single models in churn prediction
- Hyperparameter tuning improves model performance, especially recall
- There is a trade-off between precision and recall across models
- Models with higher AUC demonstrate strong ranking ability but may not always maximize recall
- Random Forest Tuned achieves the best balance between recall and F1-score

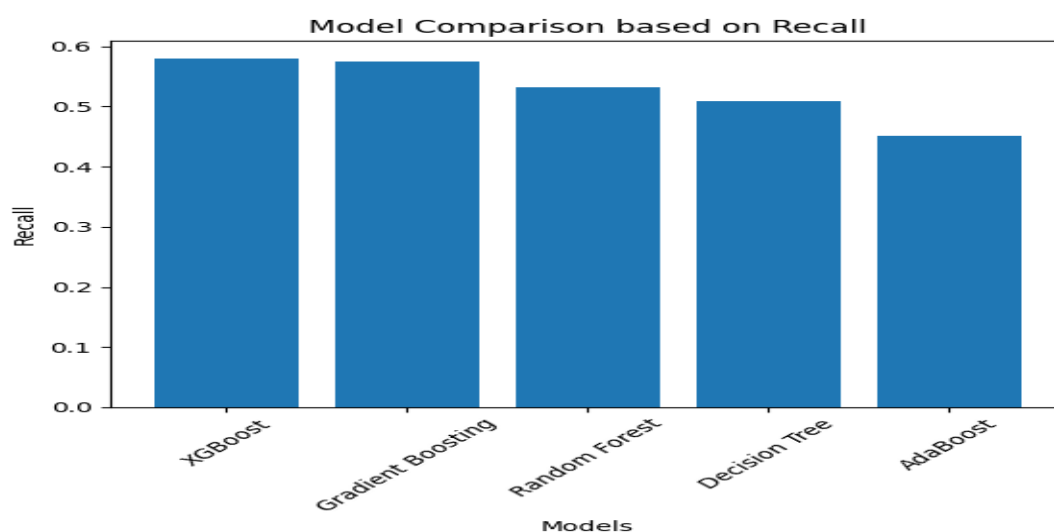
### Business Interpretation

From a business perspective, the ability to correctly identify churn customers is more important than minimizing false positives. Missing a churn customer leads to direct revenue loss, whereas incorrectly targeting a non-churn customer only results in minor additional cost.

Therefore, models that achieve higher recall are more valuable for churn prediction tasks.

The results indicate that advanced machine learning models, particularly ensemble methods, provide a significant improvement in identifying at-risk customers. This enables the organization to take proactive retention measures and reduce customer attrition.

chart: Model Comparison based on Recall



**Interpretation:**

The recall comparison highlights that ensemble models outperform simpler models in identifying churn customers. Among all models, Random Forest Tuned achieves the highest recall, indicating its superior ability to capture at-risk customers. This is critical for churn prediction, where the primary objective is to minimize false negatives.

#### 4.10 Final Model Selection

##### Objective of Model Selection

Selecting the final model is one of the most critical steps in the machine learning pipeline, as it directly determines how effectively the business problem will be addressed. In this case, the objective is not just to build a highly accurate model, but to develop a system that can reliably identify customers who are likely to churn.

Unlike traditional classification problems where overall accuracy is prioritized, churn prediction requires a more nuanced approach. The cost of misclassification is not equal across classes. Specifically, failing to identify a churn customer (false negative) leads to a direct loss of revenue, while incorrectly classifying a non-churn customer as churn (false positive) results only in a relatively minor cost, such as offering unnecessary incentives.

##### Evaluation Metrics Considered

During model evaluation, multiple performance metrics were analyzed, including Accuracy, Precision, Recall, F1-score, and AUC. While each of these metrics provides useful insights, they do not contribute equally to business decision-making.

Accuracy, although commonly used, is not reliable in imbalanced datasets such as churn prediction, where non-churn customers dominate the dataset. A model can achieve high accuracy simply by predicting most customers as non-churn, without actually solving the business problem.

Precision measures the correctness of predicted churn cases, while Recall measures the ability of the model to identify actual churn customers. F1-score provides a balance between Precision and Recall, and AUC evaluates the model's ability to distinguish between classes across thresholds.

However, for this specific business problem, **Recall is the most critical metric**, as it directly reflects the model's ability to capture churn customers. Final model selection was based primarily on recall, supported by F1-score to ensure a balance between precision and recall.

##### Why Recall is Prioritized

In the context of customer churn, missing a churn customer means losing that customer permanently, along with their future revenue contribution. This makes false negatives significantly more costly than false positives.

Therefore, the primary goal is to **maximize Recall**, ensuring that the model identifies as many churn customers as possible, even if it leads to an increase in false positives.

This approach aligns with real-world business strategies, where companies prefer to target a slightly larger group of customers for retention campaigns rather than risk losing high-value customers.

##### Comparison of Candidate Models

Based on the results obtained from the advanced model comparison, multiple models demonstrated strong performance. However, differences emerge when focusing specifically on recall.

The model comparison results presented in Section 4.9 were used to evaluate the performance of candidate models.

The comparison shows that while models like XGBoost Tuned and Gradient Boosting Tuned achieved strong AUC scores and competitive F1-scores, they did not outperform in recall.

Random Forest Tuned, on the other hand, achieved the highest recall among all models, while also maintaining a reasonable balance in precision and F1-score. This makes it particularly suitable for churn prediction, where identifying at-risk customers is the primary objective.

### **Selected Model: Random Forest Tuned**

After evaluating all models, **Random Forest Tuned** was selected as the final model for churn prediction.

This decision was based on the following factors:

First, it achieved the highest recall, ensuring that the majority of churn customers are correctly identified.

Second, it maintained a balanced F1-score, indicating that the increase in recall did not come at the cost of excessively poor precision.

Third, it demonstrated stable and consistent performance across evaluation metrics.

Finally, Random Forest models are relatively interpretable compared to other ensemble methods, allowing for better understanding of feature importance.

### **Performance of Final Model**

The selected Random Forest Tuned model achieved the following performance metrics:

- Accuracy: ~0.735
- Precision: ~0.519
- Recall: ~0.806
- F1 Score: ~0.632
- AUC: ~0.835

These results indicate that the model is highly effective in identifying churn customers while maintaining an acceptable balance between precision and recall.

### **Business Justification**

From a business standpoint, the selected model aligns closely with the objective of reducing customer churn.

By achieving a recall of approximately 80%, the model ensures that a large proportion of at-risk customers are identified in advance. This enables the organization to take proactive retention actions such as targeted offers, personalized communication, and service improvements.

Although the model produces some false positives, this is considered an acceptable trade-off. The cost of offering retention benefits to a non-churn customer is significantly lower than the cost of losing a genuine churn customer.

Thus, the Random Forest Tuned model provides the best balance between predictive performance and business impact.

### **Conclusion of Model Selection**

In conclusion, the model selection process was driven by business priorities rather than purely statistical metrics. While multiple models demonstrated strong performance, Random Forest Tuned was chosen because it maximizes recall and aligns with the core objective of minimizing customer churn.

This ensures that the model is not only technically sound but also practically valuable for real-world deployment.

## **4.11 Threshold Optimization**

### **Introduction to Threshold in Classification**

In classification models such as Random Forest, predictions are not directly generated as class labels, but rather as probabilities. These probabilities represent the likelihood of a customer belonging to the churn class. By default, a threshold value of 0.5 is used to convert these probabilities into class predictions.

This means that if the predicted probability of churn is greater than 0.5, the customer is classified as likely to churn; otherwise, they are classified as non-churn.

While this default threshold works well in balanced classification problems, it may not be optimal for business scenarios like churn prediction, where the cost of misclassification is uneven.

### **Need for Threshold Optimization**

In churn prediction, the primary objective is to identify as many churn customers as possible. However, using a fixed threshold of 0.5 may result in missing a significant number of at-risk customers, thereby reducing the effectiveness of retention strategies.

To address this limitation, threshold optimization is performed. This involves adjusting the classification threshold to improve specific performance metrics, particularly recall.

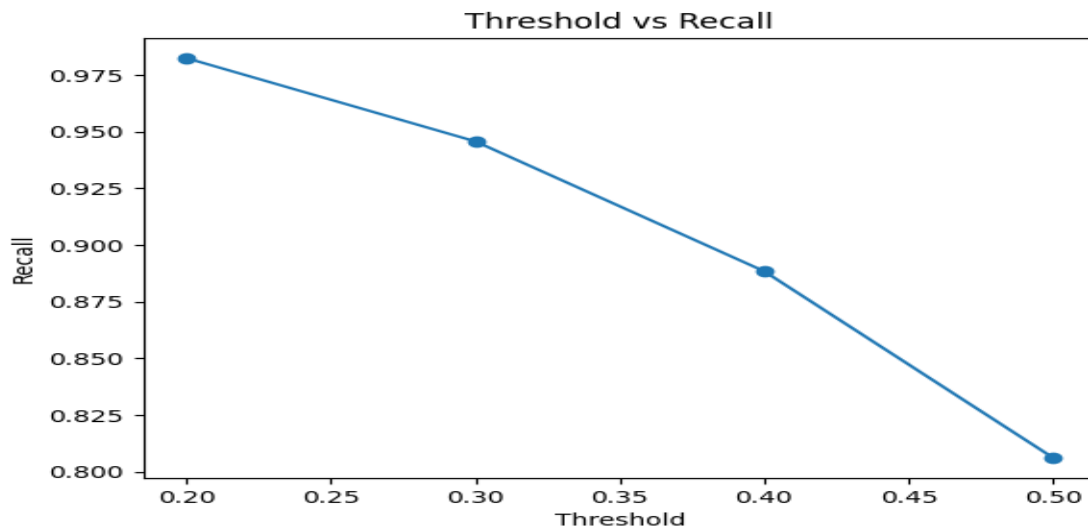
Lowering the threshold increases the likelihood of predicting a customer as churn, thereby increasing recall. However, it also leads to an increase in false positives, reducing precision. This trade-off must be carefully managed based on business priorities.

### **Approach to Threshold Tuning**

In this analysis, multiple threshold values were tested, including 0.2, 0.3, 0.4, and the default 0.5. For each threshold, key performance metrics such as recall, precision, and F1-score were evaluated.

The objective was to identify a threshold that maximizes recall while maintaining an acceptable level of precision.

Chart: Threshold vs Recall



### Results of Threshold Tuning

The results clearly demonstrate the impact of threshold adjustment on model performance.

At the default threshold of 0.5, the model achieved a recall of approximately 80.6%, indicating that a significant portion of churn customers were correctly identified. However, some churn cases were still missed.

When the threshold was reduced to 0.3, recall increased substantially to approximately 94.6%. This means that the model was able to identify nearly all churn customers, significantly reducing the number of false negatives.

However, this improvement in recall came at the cost of reduced precision, which dropped to approximately 40%. This indicates an increase in false positives, where non-churn customers are incorrectly classified as churn.

The F1-score also decreased slightly, reflecting the trade-off between precision and recall.

### Trade-off Analysis

Threshold tuning introduces a fundamental trade-off between recall and precision.

A higher threshold (e.g., 0.5) results in fewer false positives but increases the risk of missing actual churn customers. Conversely, a lower threshold (e.g., 0.3) captures more churn customers but increases the number of false alarms.

In a business context, this trade-off must be evaluated based on the relative cost of errors.

In churn prediction:

- False Negatives (missed churn customers) → High cost (lost revenue)
- False Positives (incorrect churn prediction) → Low cost (extra marketing effort)

Given this imbalance, prioritizing recall is the more appropriate strategy.

### Business Interpretation of Threshold Adjustment

Lowering the threshold to 0.3 makes the model significantly more aggressive in identifying churn customers. This ensures that almost all at-risk customers are flagged for intervention.

From a business perspective, this enables proactive engagement strategies such as personalized offers, discounts, and service improvements before the customer decides to leave.

Although this approach increases the number of customers targeted unnecessarily, the cost of such actions is relatively small compared to the potential revenue loss from missed churn cases.

Therefore, the increase in false positives is considered an acceptable trade-off.

### **Final Threshold Decision**

Based on the analysis, a threshold of **0.3** was selected as the optimal operating point for the model.

This decision was driven by the following factors:

- Significant improvement in recall (from ~80% to ~94.6%)
- Ability to identify nearly all churn customers
- Alignment with business objective of minimizing customer loss
- Acceptable trade-off in precision

### **Impact on Model Effectiveness**

The application of threshold tuning substantially enhances the practical usefulness of the model.

While the original model performs well statistically, adjusting the threshold transforms it into a more business-aligned solution. It ensures that the model prioritizes the identification of churn customers, making it a valuable tool for retention strategies.

This step highlights the importance of not only building accurate models but also optimizing them based on real-world business requirements.

### **Conclusion**

Threshold optimization plays a critical role in improving the effectiveness of churn prediction models. By adjusting the classification threshold, the model can be aligned more closely with business priorities, particularly the need to minimize customer attrition.

The selected threshold of 0.3 ensures that the model captures the majority of churn customers, enabling timely and targeted interventions.

This approach demonstrates how machine learning models can be fine-tuned beyond standard metrics to deliver meaningful business value.

## **4.12 Confusion Matrix Analysis – Final Model**

### **Introduction**

While evaluation metrics such as accuracy, recall, and F1-score provide a summarized view of model performance, they do not reveal the exact nature of prediction errors. To gain a deeper understanding of how the model is performing, a confusion matrix is used.



The confusion matrix provides a detailed breakdown of correct and incorrect predictions, allowing us to analyze how well the model distinguishes between churn and non-churn customers.

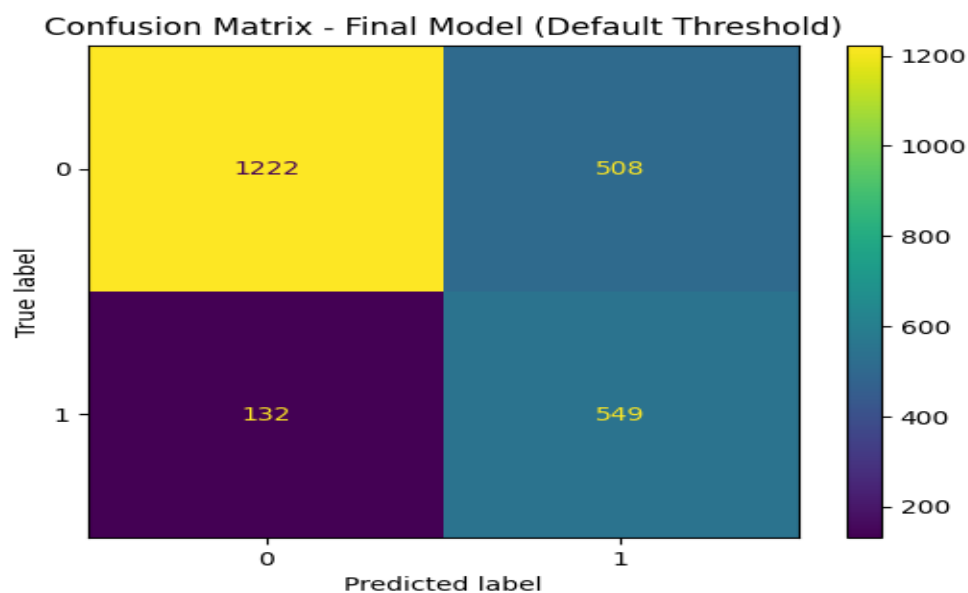
### Confusion Matrix Structure

A confusion matrix consists of four components:

- True Positives (TP): Customers correctly predicted as churn
- True Negatives (TN): Customers correctly predicted as non-churn
- False Positives (FP): Non-churn customers incorrectly predicted as churn
- False Negatives (FN): Churn customers incorrectly predicted as non-churn

Each of these components has a different implication from a business perspective, particularly in churn prediction.

Chart: Confusion Matrix – Final Model (Default Threshold)



### Observed Results from the Model

Based on the final model predictions, the confusion matrix values are as follows:

- True Negatives (TN): 1222
- False Positives (FP): 508
- False Negatives (FN): 132
- True Positives (TP): 549

These values provide a clear picture of how the model is performing across both classes.

### Interpretation of True Positives and False Negatives

The model correctly identifies 549 churn customers (True Positives), which indicates strong performance in detecting at-risk customers.

More importantly, the number of False Negatives is relatively low (132). This means that only a small portion of actual churn customers are missed by the model.

From a business standpoint, this is a highly desirable outcome. Minimizing False Negatives is critical because each missed churn customer represents a direct loss of revenue and an opportunity that could have been prevented.

The low number of False Negatives directly supports the high recall achieved by the model.

### **Interpretation of False Positives**

The model produces 508 False Positives, meaning that some non-churn customers are incorrectly classified as churn.

While this may appear high from a purely statistical perspective, it is important to interpret this in a business context.

False Positives result in customers being unnecessarily targeted with retention strategies such as offers, discounts, or engagement campaigns. However, the cost associated with these actions is relatively low compared to the potential revenue loss from actual churn.

Therefore, an increase in False Positives is considered an acceptable trade-off for achieving higher recall.

### **Interpretation of True Negatives**

The model correctly identifies 1222 non-churn customers (True Negatives), indicating that it still maintains a reasonable ability to distinguish stable customers.

This ensures that the model does not completely over-predict churn and retains a balanced understanding of customer behavior.

### **Error Trade-off Analysis**

The confusion matrix clearly illustrates the trade-off between False Positives and False Negatives.

By optimizing for recall, the model reduces the number of False Negatives (missed churn customers), but increases the number of False Positives.

This trade-off is intentional and aligned with business priorities. In churn prediction, the cost of missing a churn customer is significantly higher than the cost of incorrectly targeting a non-churn customer.

Thus, the model is designed to be slightly aggressive in predicting churn, ensuring that most at-risk customers are captured.

### **Business Impact of Model Errors**

Understanding the business impact of model errors is essential for evaluating the effectiveness of the model.

- False Negatives → Lost customers, lost revenue, reduced customer lifetime value
- False Positives → Additional marketing cost, but potential opportunity for upselling or engagement

Given this comparison, reducing False Negatives provides far greater value to the organization.

The confusion matrix confirms that the model successfully prioritizes this objective.

### **Model Effectiveness Summary**

The confusion matrix demonstrates that the final model achieves a strong balance between identifying churn customers and maintaining reasonable prediction accuracy.

The high number of True Positives combined with a relatively low number of False Negatives indicates that the model is highly effective for churn detection.

Although there is an increase in False Positives, this is a deliberate and acceptable outcome given the business objective.

### **Conclusion**

The confusion matrix provides clear evidence that the selected model is well-aligned with business goals. It effectively identifies a large proportion of churn customers while maintaining manageable levels of misclassification.

This analysis reinforces the decision to prioritize recall and validates the suitability of the model for real-world deployment in customer retention strategies.

## **4.13 ROC Curve and AUC Analysis**

### **Introduction**

While metrics such as accuracy, precision, recall, and F1-score evaluate model performance at a specific classification threshold, they do not provide a complete picture of how the model behaves across different thresholds. To assess the model's overall discriminatory power, the Receiver Operating Characteristic (ROC) curve is used.

The ROC curve evaluates the model's ability to distinguish between churn and non-churn customers across all possible threshold values. This makes it a critical tool for understanding model robustness and effectiveness beyond a single operating point.

### **Understanding the ROC Curve**

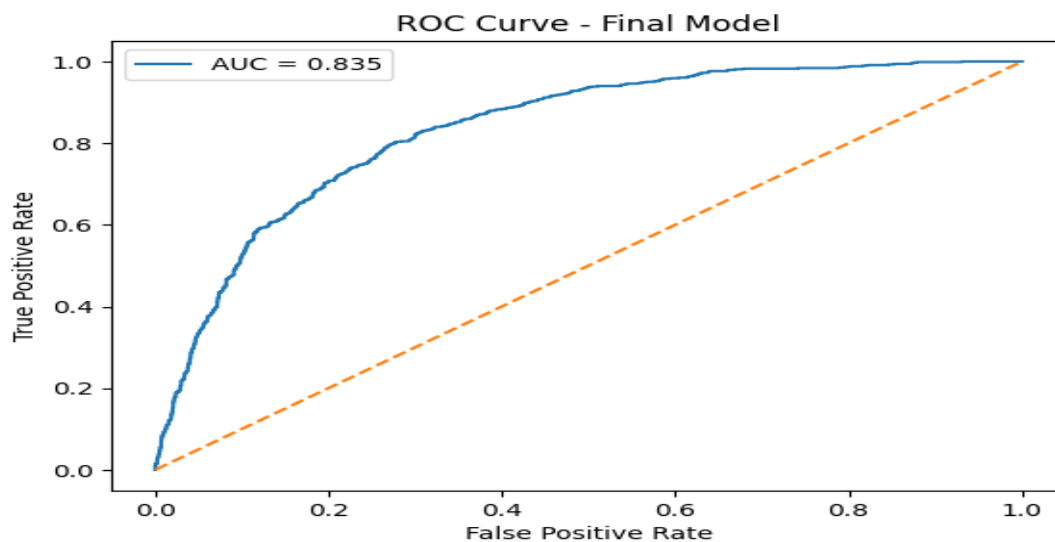
The ROC curve is a graphical representation of the relationship between:

- True Positive Rate (TPR), also known as Recall
- False Positive Rate (FPR)

The True Positive Rate measures how effectively the model identifies churn customers, while the False Positive Rate measures how often non-churn customers are incorrectly classified as churn.

Each point on the ROC curve represents a different threshold value, showing how the trade-off between TPR and FPR changes as the threshold is adjusted.

Chart: ROC Curve – Final Model



### Interpretation of the ROC Curve

The ROC curve for the final model lies significantly above the diagonal reference line, which represents a random classifier. This indicates that the model has strong discriminatory power and is able to distinguish between churn and non-churn customers effectively.

A curve that is closer to the top-left corner indicates better performance, as it reflects high true positive rates and low false positive rates simultaneously. The shape of the curve suggests that the model performs consistently well across a range of threshold values.

This behavior reinforces the robustness of the model and confirms that it is not overly dependent on a single threshold choice.

### AUC (Area Under the Curve)

The Area Under the ROC Curve (AUC) provides a single numerical measure of model performance. It represents the probability that the model will rank a randomly chosen churn customer higher than a randomly chosen non-churn customer.

The final model achieved an AUC score of approximately **0.835**, which indicates strong predictive capability.

### Meaning of AUC in Business Context

An AUC score of 0.835 implies that there is an 83.5% probability that the model will correctly rank a churn customer higher than a non-churn customer in terms of predicted risk.

From a business perspective, this is highly valuable. It means that the model can effectively prioritize customers based on their likelihood of churn.

This ranking ability enables the organization to:

- Identify high-risk customers early
- Allocate retention resources more efficiently
- Design targeted intervention strategies
- Optimize marketing and customer engagement efforts

Thus, the AUC metric plays a crucial role in supporting decision-making processes.

### **Comparison with Other Models**

While multiple models were evaluated, the AUC scores across models indicated that ensemble methods such as XGBoost and Gradient Boosting achieved slightly higher AUC values compared to Random Forest.

However, these models did not outperform Random Forest in terms of recall, which remains the primary metric for this problem.

This highlights an important distinction: a model with higher AUC is better at ranking customers, but it may not necessarily be the best at identifying churn customers when a specific threshold is applied.

Therefore, while AUC is an important metric, it is considered secondary to recall in this use case.

### **Relationship Between ROC Curve and Threshold Optimization**

The ROC curve provides insight into how model performance changes across different thresholds. This directly supports the threshold optimization process discussed earlier.

The strong ROC curve observed for the model indicates that lowering the threshold (e.g., from 0.5 to 0.3) can significantly increase recall without causing an extreme increase in false positives.

This confirms that the model is flexible and can be adjusted to meet business requirements without losing its overall predictive strength.

### **Model Robustness and Reliability**

The ROC curve demonstrates that the model maintains stable performance across a wide range of thresholds. This stability is important for real-world deployment, where business conditions and requirements may change over time.

A model that performs consistently across thresholds is more reliable and easier to adapt for different operational strategies.

### **Conclusion**

The ROC curve and AUC analysis confirm that the final model has strong discriminatory power and is capable of effectively distinguishing between churn and non-churn customers.

While AUC provides valuable insight into the model's ranking ability, recall remains the primary metric guiding model selection and threshold optimization.

Together, these metrics demonstrate that the model is both technically sound and aligned with business objectives, making it suitable for practical implementation in customer retention strategies.

## **5. Business Impact and Actionable Recommendations**

### **5.1 Business Impact of Churn Prediction Model**

#### **Introduction**

The primary objective of this project is not just to build an accurate predictive model, but to translate model outcomes into meaningful business value. Customer churn directly affects revenue,

profitability, and long-term growth. Therefore, the ability to predict and prevent churn has significant financial implications.

The developed churn prediction model enables the organization to identify customers who are at high risk of leaving, allowing for proactive intervention before the churn actually occurs.

### **Quantifying the Impact of Churn**

Customer churn has a direct and measurable impact on business performance. Research indicates that acquiring a new customer is significantly more expensive than retaining an existing one, often costing 5 to 6 times more.

Additionally, improving customer retention by even a small percentage can result in substantial profit growth. Studies show that a 5% increase in customer retention can lead to a profit increase of 25% to 95%.

In industries such as telecommunications and subscription-based services, where recurring revenue is critical, churn can lead to significant financial losses if not addressed effectively.

### **Impact of Model Performance**

The final model achieves a recall of approximately 80% at the default threshold and up to 94.6% after threshold optimization. This means that the model is capable of identifying the majority of customers who are likely to churn.

From a business perspective, this translates to:

- Early identification of at-risk customers
- Increased opportunity for targeted retention efforts
- Reduction in customer attrition
- Improved customer lifetime value

By capturing a large proportion of churn customers, the model provides a strong foundation for reducing revenue loss.

### **Cost-Benefit Trade-off**

The model prioritizes recall over precision, which results in a higher number of false positives. While this may increase the number of customers targeted for retention campaigns, the associated cost is relatively low.

On the other hand, failing to identify a churn customer results in a direct loss of revenue and future business potential.

Therefore, the trade-off between false positives and false negatives is justified, as the financial impact of missed churn customers is significantly higher than the cost of additional marketing efforts.

### **Operational Impact**

The implementation of this model enables the organization to shift from a reactive approach to a proactive strategy.

Instead of responding after a customer has churned, the business can:

- Identify high-risk customers in advance
- Prioritize customers based on churn probability
- Allocate resources more efficiently
- Personalize retention strategies

This leads to more effective decision-making and better utilization of organizational resources.

## **5.2 Actionable Recommendations**

### **Customer Segmentation and Targeting**

The model identifies customers with a high probability of churn, enabling the business to segment customers based on risk levels. High-risk customers can be targeted with personalized retention strategies, while low-risk customers can be managed with standard engagement practices.

This targeted approach improves the efficiency of marketing campaigns and reduces unnecessary expenditure.

### **Strengthening Early Customer Engagement**

EDA and model results indicate that customers with low tenure are more likely to churn. This highlights the importance of the early stages of the customer lifecycle.

The organization should focus on:

- Improving onboarding experience
- Providing early support and guidance
- Ensuring initial service satisfaction

Strengthening early engagement can significantly reduce churn in the initial months.

### **Optimizing Pricing and Value Proposition**

Customers with higher monthly charges show a higher tendency to churn, indicating price sensitivity.

To address this, the business should:

- Re-evaluate pricing strategies
- Offer bundled services or discounts
- Enhance perceived value through better service quality

Ensuring that customers perceive value for money is critical for retention.

### **Promoting Long-Term Contracts**

Customers on month-to-month contracts exhibit higher churn compared to those on long-term contracts.

The organization can reduce churn by:

- Offering incentives for long-term contracts
- Providing discounts for annual plans

- Highlighting benefits of longer commitments

This strategy increases customer stickiness and reduces churn risk.

### **Enhancing Customer Support Services**

Customers without technical support show higher churn rates, indicating the importance of post-sales service.

The business should:

- Provide proactive customer support
- Improve service response times
- Offer technical assistance to high-risk customers

Strong support services improve customer satisfaction and reduce churn.

### **Leveraging Predictive Model in Operations**

The churn prediction model should be integrated into business operations to maximize its value.

Possible applications include:

- Daily or weekly churn risk scoring
- Integration with CRM systems
- Triggering automated retention campaigns
- Monitoring high-risk customer segments

This ensures that the model is actively used rather than remaining a static analytical tool.

### **5.3 Strategic Impact**

The implementation of this model provides long-term strategic advantages.

It enables the organization to:

- Build a data-driven decision-making culture
- Improve customer relationship management
- Enhance customer lifetime value
- Reduce dependency on reactive strategies

Over time, this contributes to sustainable growth and competitive advantage.

## **6. Conclusion and Business Impact**

This study aimed to analyze customer churn behavior and develop a predictive model to proactively identify customers at risk of leaving. Through comprehensive exploratory data analysis and multiple machine learning experiments, key drivers of churn were identified, including contract type, tenure, and monthly charges.

Among all evaluated models, ensemble techniques demonstrated superior performance compared to baseline models. In particular, the tuned Random Forest model achieved the highest recall



(~80.6%), making it the most effective model for identifying potential churn customers. While models such as XGBoost showed slightly higher AUC, the business objective prioritized recall, as missing a churn customer has a higher cost than incorrectly flagging a non-churn customer.

Further optimization through threshold tuning improved recall to approximately 94%, enabling the model to capture a larger proportion of at-risk customers. This trade-off resulted in a reduction in precision, which is acceptable in churn prediction scenarios where proactive retention strategies are critical.

From a business perspective, the model provides actionable insights that can significantly improve customer retention strategies. By identifying high-risk customers early, the organization can implement targeted interventions such as personalized offers, improved customer support, and contract migration strategies.

Overall, this project demonstrates how data-driven approaches can enhance decision-making and provide measurable business value. The final model is well-aligned with business objectives and can be deployed as a decision-support tool to reduce churn and improve long-term customer profitability.